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Online adaptive Wiener filter for channel denoising in MIMO-OFDM systems

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ABSTRACT

In multiple-input multiple-output (MIMO) orthogonal frequency-division multiplexing (OFDM) systems, channel estimation errors are unavoidable due to the limited availability of reference signals. To address this, we propose an online adaptive denoising method based on the Wiener filtering principle. The proposed filter is optimized using an online training dataset constructed from noisy channel estimates, with high-quality labels generated through data-aided channel estimation to compensate for the absence of true channel information. This approach enables adaptive filter design that dynamically responds to varying channel conditions. Simulation results confirm that the proposed method substantially enhances the performance of conventional channel estimation.

1. Introduction

Modern wireless communication systems demand high reliability, high capacity, and low latency [1]. To meet these demands, multiple-input multiple-output (MIMO) technology and orthogonal frequency-division multiplexing (OFDM) have been widely adopted. MIMO enhances spectral efficiency by utilizing multiple transmit and receive antennas, whereas OFDM mitigates inter-symbol interference caused by multipath propagation [1,2]. The integration of these technologies, MIMO-OFDM, has become fundamental to modern communication standards such as 5G NR [3].

Accurate channel state information at the receiver (CSIR) is crucial for realizing the full potential of MIMO-OFDM systems. In practice, CSIR is typically estimated using demodulation-reference signals (DM-RS), which are sparsely distributed across the time-frequency grid [4]. Linear estimators, such as least squares (LS) and linear minimum mean square error (LMMSE), are commonly employed to estimate the channel at DM-RS positions, followed by interpolation to recover the channel information at data-bearing positions [5]. However, the sparse placement of DM-RSs and the presence of channel noise inevitably introduce estimation errors at the pilot locations, which are further propagated through the interpolation process, leading to a degradation in overall system performance [6]. This challenge motivates the development of advanced channel denoising techniques aimed at refining the channel estimates after the initial estimation step.

Channel denoising has been considered a practical yet effective technique to mitigate channel estimation errors in communication systems. A key advantage of the channel denoising lies in the fact that it requires no repetition of channel estimation, data detection, and channel decoding processes; rather, the channel denoising is applied only once after the channel estimation process in order to *explicitly* reduce the errors in channel estimation outputs (i.e., noisy channel estimates).

Recent studies have explored channel denoising techniques using adaptive filtering and data-driven approaches. For instance, [7] proposed a sparsity-aware denoising algorithm for underwater acoustic OFDM systems, where pilot-based estimates are refined through adaptive filtering. [8] developed a reinforcement learning-based framework for successive channel denoising in MIMO-OFDM systems. [9] applied deep neural networks trained offline to improve channel frequency response (CFR) estimates in conventional OFDM settings. While these methods show promising performance under specific assumptions, they often rely on prior channel models, offline training, or static estimation frameworks, which may limit their applicability to dynamic and unpredictable wireless environments. In contrast, the method proposed in this work performs denoising in a fully online and model-free manner. It dynamically adapts to the current channel condition using only noisy CFR observations collected during communication, without requiring any statistical priors or pre-trained models.

In this work, we propose a data-driven Wiener-filter-based denoising method for online adaptive channel denoising in MIMO-OFDM systems. The core idea is to adapt the Wiener filter taps during communication using only observed noisy channel estimates. To enable supervised training, we generate an online training dataset where the

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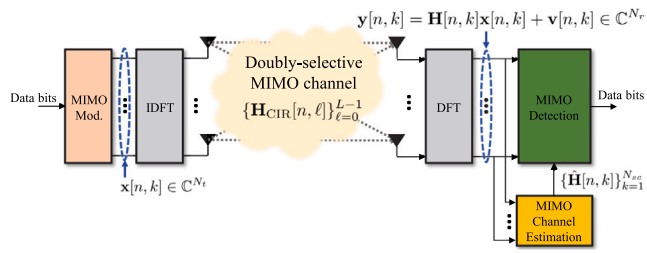

Fig. 1. An illustration of a MIMO-OFDM communication system with doubly-selective channel impulse response $\mathbf{H}_{\mathrm{CIR}}[n,l]$.

input samples are noisy channel estimates, and the labels are high-quality estimates generated via data-aided channel estimation. This method treats detected data symbols as virtual pilots to refine the channel estimates without requiring ground truth. The proposed method does not require any prior knowledge of channel or noise statistics and avoids explicit statistical modeling. Instead, it learns to adapt to the current channel environment based on online-observed data. Simulation results confirm that our approach significantly reduces channel estimation errors and improves the performance of conventional estimators in various realistic channel scenarios.

2. System model

We consider a MIMO-OFDM communication system, illustrated in Fig. 1, where a transmitter equipped with N_t antennas communicates with a receiver equipped with N_r antennas over K consecutive subcarriers. The fundamental resource unit is the resource element (RE), defined by one subcarrier in the frequency domain and one OFDM symbol in the time domain.

The wireless channel is modeled as a time-varying finite impulse response filter with L taps, where L corresponds to the maximum delay spread of the channel. Specifically, the channel impulse response is assumed to follow a multivariate, time-varying model based on the cluster delay line (CDL) model standardized by 3GPP [10]. Given this model, the CFR at the (n,k)th RE can be expressed as

$$\mathbf{H}[n,k] = \sum_{l=0}^{L-1} \alpha_l[n] e^{-j2\pi k \Delta_f \tau_l}, \tag{1}$$

where Δ_f is the subcarrier spacing, $\alpha_l[n]$ denotes the complex gain of the lth multipath component at time index n, and τ_l represents the delay of the lth path.

(1) **DM-RS Transmission:** To enable coherent demodulation at the receiver, a subset of REs is reserved for the transmission of DM-RSs, following the 5G NR standard. The DM-RSs are pre-known at both the transmitter and receiver and serve as pilots for channel estimation. Let $\mathcal{I}_p \subset \{(n,k)\}_{\forall n,k}$ denote the set of REs indices allocated for DM-RS transmission. The transmitted DM-RS at (n,k)th RE is represented by $\mathbf{x}_p[n,k] = [x_{p,1}[n,k], \ldots, x_{p,N_t}[n,k]]^{\mathsf{T}} \in \mathbb{C}^{N_t}$, where each symbol satisfies the power normalization $\mathbb{E}[|x_{p,i}[n,k]|^2] = 1$. Accordingly, for $(n,k) \in \mathcal{I}_p$, the received signal at the DM-RS positions can be written as

$$\mathbf{y}[n,k] = \mathbf{H}[n,k]\mathbf{x}_p[n,k] + \mathbf{v}[n,k], \tag{2}$$

where $\mathbf{v}[n,k] \sim \mathcal{CN}\left(\mathbf{0}_{N_r}, \sigma^2 \mathbf{I}_{N_r}\right)$ is an independent circularly symmetric complex Gaussian noise vector and σ^2 is a noise variance.

(2) **Data Transmission:** The REs not assigned to DM-RSs, indexed by $\mathcal{I}_d \subset \{(n,k)\}_{\forall n,k}$, are utilized for data transmission. The transmitted data symbol at the (n,k)th RE is denoted as $\mathbf{x}[n,k] = [x_1[n,k], \ldots, x_{N_t}[n,k]]^{\mathsf{T}} \in \mathcal{X}^{N_t}$, where $\mathcal{X}$ is the modulation constellation set (e.g., QPSK, 16-QAM), and each symbol satisfies $\mathbb{E}[|x_i[n,k]|^2] = 1$. Then, for $(n,k) \in \mathcal{I}_p$, the received signal at the (n,k)th RE is modeled as

$$\mathbf{y}[n,k] = \mathbf{H}[n,k]\mathbf{x}[n,k] + \mathbf{v}[n,k]. \tag{3}$$

The main objective at the receiver is to accurately recover $\mathbf{x}[n,k]$ from $\mathbf{y}[n,k]$ using channel estimates $\hat{\mathbf{H}}[n,k]$.

(3) **Channel Estimation:** Channel estimation in practical MIMO-OFDM systems typically proceeds in two stages. First, the CFRs at the DM-RS positions are estimated using the known pilot symbols. Second, interpolation techniques are applied to infer the CFRs at non-DM-RS positions. Formally, we denote by $\hat{\mathbf{H}}^{(i,j)} = \hat{\mathbf{H}}[n_i, k_j]$ the estimated CFR at the (i,j)th DM-RS location in the time–frequency domain, where n_i and k_j are the corresponding OFDM symbol and subcarrier indices.

Linear estimators such as the LS and LMMSE methods are commonly used due to their low complexity and satisfactory performance in high signal-to-noise ratio (SNR) conditions. However, in low-SNR regimes, the sparse deployment of DM-RSs provides insufficient noise averaging, causing severe degradation in channel estimation quality. This degradation is further exacerbated during interpolation, as noisy CFR estimates at pilot locations lead to error propagation across data REs. Consequently, the detection performance and the reliability of the communication link are significantly impacted.

3. Proposed online adaptive Wiener filter for channel denoising

In this section, we present an online adaptive Wiener filter for channel denoising in MIMO-OFDM systems, aiming to alleviate errors in the CFR estimates in the DM-RS positions based on the Wiener filtering principle [11].

3.1. Basic idea: Denoising filter

The core idea of our approach is to apply a causal denoising filter that combines the current noisy channel frequency response (CFR) estimate with preceding estimates within a predefined time–frequency window, as illustrated in Fig. 2. The input vector to the filter, denoted by $\hat{\mathbf{h}}_{r,t}^{(i,j)}$, is constructed by collecting noisy CFR values from a local window centered at the DM-RS resource element (n_i, k_j), arranging them into a two-dimensional matrix indexed by frequency and time, and then vectorizing them in column-wise order:

$$\hat{\mathbf{h}}_{r,t}^{(i,j)} = \left[\hat{h}_{r,t}^{(i-W_t+1, j-W_f+1)}, \ldots, \hat{h}_{r,t}^{(i,j)}\right]^{\mathsf{T}} \in \mathbb{C}^{W_t W_f \times 1}.$$

The vector length $W_t \times W_f$ determines the input size of the denoising filter.

The corresponding output of the filter is then given by

$$\tilde{h}_{r,t}^{(i,j)} = f_{\mathrm{filter}}\left(\hat{\mathbf{h}}_{r,t}^{(i,j)}\right). \tag{4}$$

The underlying intuition is to exploit the temporal and frequency-domain correlations among neighboring CFRs in order to reduce the estimation error at the current position. This strategy is particularly effective in practical MIMO-OFDM systems, where CFRs within a small time–frequency neighborhood tend to exhibit strong correlation. To maximize the denoising performance, it is essential to optimize the filter taps based on the relationship between noisy CFR estimates and their corresponding ground-truth values. In the following subsections, we describe a practical method for learning this relationship and discuss how to optimize the filter accordingly.

3.2. Wiener filter for channel denoising

To design an effective denoising filter, we adopt the Wiener filtering principle [11], which is well known for its optimality in minimizing the MSE between the filtered output and the desired signal. According to this principle, we aim to find the optimal linear filter $\mathbf{a}^\star = [a_1^\star, a_2^\star, \ldots, a_{W_t W_f}^\star]$ that minimizes the MSE between the filtered noisy CFR estimate $\mathbf{a}^\star \hat{\mathbf{h}}_{r,t}^{(i,j)}$ and the corresponding denoised target CFR $\tilde{h}_{r,t}^{(i,j)}$. We assume that the true CFR at the DM-RS positions forms a wide-sense stationary (WSS) two-dimensional stochastic process with zero mean, and that the noisy CFR estimate is a corrupted observation of

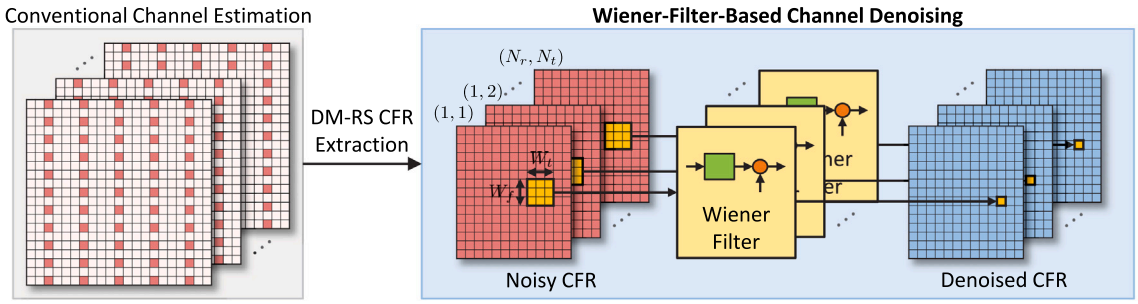

Fig. 2. An illustration of the proposed Wiener-filter-based channel denoising process.

the true CFR affected by additive white Gaussian noise. Under these assumptions, the optimal filter $\mathbf{a}^\star$ is obtained by solving the following MSE minimization problem:

$$\mathbf{a}^\star = \underset{\mathbf{a}\in\mathbb{C}^{W_tW_f}}{\operatorname{argmin}}\ \mathbb{E}\left[\left|\tilde{h}_{r,t}^{(i,j)} - \mathbf{a}\hat{\mathbf{h}}_{r,t}^{(i,j)}\right|^2\right]. \tag{5}$$

For the remainder of this subsection, we omit the indices r, t, i, and j for notational simplicity, as the same denoising process is independently applied to all elements in the set $\{\hat{h}_{r,t}^{(i,j)}\}_{r,t,i,j}$.

Using the orthogonality principle, the optimal filter coefficients $\{a_u^\star\}$ can be obtained by solving the Wiener–Hopf equations [11]:

$$\mathbb{E}[\tilde{h}\,(\hat{\mathbf{h}})_u^*] = \sum_v a_v^\star\,\mathbb{E}\left[(\hat{\mathbf{h}})_v\,(\hat{\mathbf{h}})_u^*\right], \tag{6}$$

where $(\hat{\mathbf{h}})_u$ denotes the uth entry of the vector $\hat{\mathbf{h}}$. The solution to these equations is given in closed form as

$$\mathbf{a}^\star = \boldsymbol{\theta}\boldsymbol{\Phi}^{-1}, \tag{7}$$

where $\boldsymbol{\theta} = \mathbb{E}[\tilde{h}\,\hat{\mathbf{h}}^{\mathsf{H}}] \in \mathbb{C}^{1\times W_tW_f}$ is the cross-covariance vector between the target and the noisy CFR estimates, and $\boldsymbol{\Phi} = \mathbb{E}[\hat{\mathbf{h}}\,\hat{\mathbf{h}}^{\mathsf{H}}] \in \mathbb{C}^{W_tW_f\times W_tW_f}$ is the autocovariance matrix of the noisy estimates. Exact computation of the optimal filter $\mathbf{a}^\star$ requires knowledge of both $\boldsymbol{\theta}$ and $\boldsymbol{\Phi}$. However, obtaining these quantities is generally infeasible in online communication settings, as it requires full knowledge of the correlation between the target CFR and the estimation error, as well as the time–frequency correlation structure among CFRs at DM-RS positions.

3.3. Online training data for filter design

To enable the optimization of the Wiener filter, we develop a practical strategy for generating online training data samples. A key challenge in generating these samples is that the true CFRs $\{h_{r,t}^{(i,j)}\}$ are not available at the receiver during online communication. To address this limitation, we approximate the target CFR using a *high-quality* estimate obtained via data-aided channel estimation [12]. This technique is known to reduce estimation error by treating detected data symbols as virtual DM-RSs and leveraging them in the estimation process. Inspired by this idea, we re-estimate each CFR $\hat{h}_{r,t}^{(i,j)}$ at the (i,j)th DM-RS position by incorporating adjacent detected data signals, in addition to the original DM-RS, as virtual DM-RSs. Let $\hat{\mathbf{x}}[n,k]$ denote the detected data symbol at the (n,k)th RE for $(n,k)\in\mathcal{I}_d$. We define Q_t and Q_f as the time and frequency ranges, respectively, that determine the neighborhood of adjacent detected symbols to be used for estimation.

Unfortunately, if the adjacent detected symbols contain detection errors, a direct use of these symbols as virtual DM-RSs may decrease the performance of data-aided channel estimation. To address this issue, we introduce a selection mechanism based on a posteriori probability (APP), which allows us to include only high-confidence detected symbols in the estimation process. To quantify the reliability of each detected symbol, we use the log-likelihood ratio (LLR) obtained from soft-decision demodulation [13]. Let $b_{t,l}[n,k]$ denote the lth bit transmitted from the tth antenna at the (n,k)th RE. The corresponding LLR is defined as

$$\lambda_{t,l}[n,k] = \log\frac{\mathbb{P}(b_{t,l}[n,k]=1\mid\mathbf{y}[n,k])}{\mathbb{P}(b_{t,l}[n,k]=0\mid\mathbf{y}[n,k])}. \tag{8}$$

The bit-wise APP is then derived as

$$\mathbb{P}(b_{t,l}[n,k]=1) = \frac{e^{\lambda_{t,l}[n,k]}}{1+e^{\lambda_{t,l}[n,k]}}, \tag{9}$$

$$\mathbb{P}(b_{t,l}[n,k]=0) = 1-\mathbb{P}(b_{t,l}[n,k]=1). \tag{10}$$

Using these bit-wise APPs, the APP of the detected symbol vector $\hat{\mathbf{x}}[n,k]$ approximating the transmitted symbol $\mathbf{x}[n,k]$ for $(n,k)\in\mathcal{I}_d$ is computed as

$$\begin{aligned}\alpha[n,k] &= \mathbb{P}(\mathbf{x}[n,k]=\hat{\mathbf{x}}[n,k])\\ &\approx \prod_{t=1}^{N_t}\prod_{l=1}^{\log_2 M}\mathbb{P}\big(b_{t,l}[n,k]=k_l(\hat{x}_t[n,k])\big),\end{aligned} \tag{11}$$

where M is the modulation order and $k_l(c)$ denotes the lth bit of the constellation point c. Note that we set $\alpha[n,k]=1$ for $(n,k)\in\mathcal{I}_p$.

We treat the APP in (11) as a reliability level of the detected symbols. Based on this interpretation, we select the detected symbols with APP values exceeding a reliability threshold $\tau_{\text{APP}}\in[0,1]$ as virtual DM-RSs. Let $\mathcal{I}_{Q_t,Q_f}^{(i,j)}$ denote the indices of the time-frequency window centered at the DM-RS position (n_i,k_j), defined as

$$\mathcal{I}_{Q_t,Q_f}^{(i,j)} = \left\{(n,k)\,\middle|\,|n-n_i|\le Q_t,\ |k-k_j|\le Q_f\right\}. \tag{12}$$

Utilizing this notation, the set of the selected detected symbols around the DM-RS position (n_i,k_j) is expressed as

$$\mathcal{I}_{\text{sel}}^{(i,j)} = \left\{(n,k)\,\middle|\,(n,k)\in\mathcal{I}_{Q_t,Q_f}^{(i,j)},\alpha[n,k]\ge\tau_{\text{APP}}\right\}. \tag{13}$$

Let $(n_u^{(i,j)},k_u^{(i,j)})$ be the uth entry of $\mathcal{I}_{\text{sel}}^{(i,j)}$ for $u\in\{1,\dots,Q_{i,j}\}$, where $Q_{i,j}$ is the size of $\mathcal{I}_{\text{sel}}^{(i,j)}$. Then, the virtual pilot matrix used for data-aided channel estimation is constructed as

$$\hat{\mathbf{X}}_{\text{sel}}[n_i,k_j] = \left[\hat{\mathbf{x}}\left[n_1^{(i,j)},k_1^{(i,j)}\right],\dots,\hat{\mathbf{x}}\left[n_{Q_{i,j}}^{(i,j)},k_{Q_{i,j}}^{(i,j)}\right]\right]. \tag{14}$$

For notational consistency, we define $\hat{\mathbf{x}}[n_i,k_j] = \mathbf{x}_p[n_i,k_j]$ for all $(n_i,k_j)\in\mathcal{I}_p$.

Suppose that the size $(2Q_t+1)\times(2Q_f+1)$ of the time-frequency window is sufficiently small such that the CFRs within this region are highly correlated. In this case, the received signal matrix that consists of the received signals in the time-frequency window centered at the (n_i,k_j)th RE can be approximated as

$$\mathbf{Y}_{\text{sel}}[n_i,k_j] \approx \mathbf{H}[n_i,k_j]\hat{\mathbf{X}}_{\text{sel}}[n_i,k_j] + \mathbf{V}_{\text{sel}}[n_i,k_j], \tag{15}$$

where

$$\begin{aligned}\mathbf{Y}_{\text{sel}}[n_i,k_j] &= \left[\mathbf{y}\left[n_1^{(i,j)},k_1^{(i,j)}\right],\dots,\mathbf{y}\left[n_{Q_{i,j}}^{(i,j)},k_{Q_{i,j}}^{(i,j)}\right]\right],\\ \mathbf{V}_{\text{sel}}[n_i,k_j] &= \left[\mathbf{v}\left[n_1^{(i,j)},k_1^{(i,j)}\right],\dots,\mathbf{v}\left[n_{Q_{i,j}}^{(i,j)},k_{Q_{i,j}}^{(i,j)}\right]\right].\end{aligned} \tag{16}$$

Then the LMMSE-based data-aided estimate is computed as

$$\tilde{\mathbf{H}}^{(i,j)} = \mathbf{Y}[n_i, k_j]\,\hat{\mathbf{X}}^{\mathrm{H}}[n_i, k_j]\left(\hat{\mathbf{X}}[n_i, k_j]\hat{\mathbf{X}}^{\mathrm{H}}[n_i, k_j] + \sigma^2\mathbf{I}\right)^{-1}.$$

This corresponds to a special case of the LMMSE estimator under the assumption of uncorrelated Rayleigh fading, where the channel covariance matrix is set to identity.

To generate the online training data samples, we designate the first N_{train} consecutive slots (i.e., $N_{\text{OFDM}} \times N_{\text{train}}$ OFDM symbols) as the online training period, during which these samples are collected. During this phase, we treat the noisy channel estimates $\hat{\mathbf{h}}_{r,t}^{(i,j)}$ as input features and the corresponding data-aided estimates $\tilde{h}_{r,t}^{(i,j)}$ as supervised target labels. Let $\mathcal{D}_{\text{online}}$ denote the set of supervised training samples generated during this period, defined as

$$\mathcal{D}_{\text{online}} = \left\{ \left(\hat{\mathbf{h}}_{r,t}^{(i,j)}, \tilde{h}_{r,t}^{(i,j)}\right) \,\middle|\, n_i \le N_{\text{OFDM}} N_{\text{train}},\ k_j \le K \right\},$$

where $\tilde{h}_{r,t}^{(i,j)}$ is the data-aided target CFR extracted as the (r,t)th entry of $\tilde{\mathbf{H}}^{(i,j)}$, and N_{OFDM} is the number of OFDM symbols in a slot. This dataset $\mathcal{D}_{\text{online}}$ is then used to empirically compute the autocovariance $\boldsymbol{\Phi}$ and the cross-covariance $\boldsymbol{\theta}$, which are used to derive the supervised Wiener filter coefficients prior to the inference phase.

3.4. Online adaptive wiener filter

By leveraging the online dataset, we empirically estimate both the cross-covariance vector $\boldsymbol{\theta}$ and the autocovariance matrix $\boldsymbol{\Phi}$ using the online training dataset $\mathcal{D}_{\text{online}}$. In particular, the cross-covariance vector $\boldsymbol{\theta}$ is approximated as

$$\boldsymbol{\theta} \approx \frac{1}{N_{\text{online}}} \sum_{(\hat{\mathbf{h}}', \tilde{h}') \in \mathcal{D}_{\text{online}}} \tilde{h}'(\hat{\mathbf{h}}')^{\mathrm{H}}. \tag{17}$$

Similarly, the autocovariance matrix $\boldsymbol{\Phi}$ is approximated as

$$\boldsymbol{\Phi} \approx \frac{1}{N_{\text{online}}} \sum_{\hat{\mathbf{h}}' \in \mathcal{D}_{\text{online}}} \hat{\mathbf{h}}'(\hat{\mathbf{h}}')^{\mathrm{H}}. \tag{18}$$

By substituting the empirical estimates from (17) and (18) into the Wiener solution, we compute the data-driven denoising filter as

$$\hat{\mathbf{a}}^{\star} = \left(\sum_{(\hat{\mathbf{h}}', \tilde{h}') \in \mathcal{D}_{\text{online}}} \tilde{h}'(\hat{\mathbf{h}}')^{\mathrm{H}} \right) \left(\sum_{(\hat{\mathbf{h}}', \tilde{h}') \in \mathcal{D}_{\text{online}}} \hat{\mathbf{h}}'(\hat{\mathbf{h}}')^{\mathrm{H}} \right)^{-1}, \tag{19}$$

which requires no additional information other than the online training dataset. The accuracy of these empirical estimates improves as the size of $\mathcal{D}_{\text{online}}$ increases. Consequently, the proposed filter can effectively reduce the MSE between the filtered output and the true CFR when a sufficiently large number of high-quality training samples is available.

When employing the proposed online adaptive Wiener filter, the channel denoising procedure comprises two phases: the online training period and the inference period. During the online training period, the receiver constructs the online training dataset $\mathcal{D}_{\text{online}}$ and uses it to design the denoising filter following the Wiener filter formulation. Importantly, data transmission and reception take place at the same time during this phase, so the system's spectral efficiency is not affected. In the inference period, the receiver applies the trained filter to denoise the CFR estimates at the DM-RS positions. As discussed in Section 3.2, the proposed filter is causal; that is, it utilizes only past and present CFR estimates within a $(W_t \times W_f)$-sized time–frequency window. As a result, the filter cannot be applied to the first $(W_f - 1)$ DM-RS positions in the frequency domain. However, since the number of such positions is small compared to the total number of DM-RS estimates, their effect on overall performance is negligible. It is also worth noting that denoising is performed only at the DM-RS positions. The CFRs at non-DM-RS positions are then reconstructed via conventional two-dimensional interpolation based on the denoised DM-RS estimates. We also analyze the computational complexity of the proposed method by dividing it into three main components. Let $M = W_t W_f$ denote the size of the time-frequency window, and N represent the number of samples. The total cost of one update step comprises covariance estimation $\mathcal{O}(NM^2)$, Wiener filter computation $\mathcal{O}(M^3)$, and data-aided label generation $\mathcal{O}(NM^3)$. For typical parameters ($M = 9, N = 32$), this result is significantly lighter than the cost of FFT or OFDM operation. Since the update is performed only once before inference, the overall computational overhead is modest.

4. Simulation results

We evaluate the frame error ratio (FER) and normalized mean squared error (NMSE) performances of the proposed channel denoising method using Monte Carlo simulations. In these simulations, we consider a MIMO-OFDM system operating at a center frequency of 3.5 GHz with a subcarrier spacing of 15 kHz and a total of 512 subcarriers. Each slot consists of 14 OFDM symbols (i.e., $N_{\text{OFDM}} = 14$), and each resource block comprises 12 subcarriers. We assume 4-QAM modulation for symbol mapping, and the number of transmit and receive antennas is set to $(N_t, N_r) = (2, 16)$. The SNR is defined as $\text{SNR} = \frac{N_t}{\sigma^2}$, and the NMSE is defined as

$$\text{NMSE} = 10\log_{10} \frac{\mathbb{E}\left[\|\hat{\mathbf{H}}[n,k] - \mathbf{H}[n,k]\|_F^2\right]}{\mathbb{E}\left[\|\mathbf{H}[n,k]\|_F^2\right]}. \tag{20}$$

We employ two-dimensional linear interpolation to reconstruct the CFRs at non-DM-RS positions. All simulations are conducted under the CDL-B channel model, with a delay spread of 300 ns and a user velocity of 120 km/h. For channel coding, we adopt a low-density parity-check code with a code rate of 1/2.

To evaluate performance, we compare the proposed method with the following baselines: (i) *Perfect CSIR*, an ideal case in which the receiver has perfect knowledge of all CFRs. (ii) *Conventional CE*, where CFRs at DM-RS positions are estimated via LS and interpolated over the remaining REs. (iii) *Data-Aided CE*, where data detection is performed based on the conventional CE method, and then the CFRs at DM-RS positions are re-estimated using the data-aided channel estimation method. (iv) *Proposed* method (Online or True), which corresponds to the proposed online adaptive Wiener filter to denoise the CFR estimates at the DM-RS positions. For this method, the time–frequency window size is set to $(W_t, W_f) = (3, 3)$. To generate the online training dataset, data-aided channel estimation is performed using a local neighborhood defined by $(Q_t, Q_f) = (2, 2)$ in the time and frequency domains, respectively. The LMMSE method is consistently used for data-aided channel estimation and data detection. For online data generation, we set $N_{\text{train}} = 4$, with $N_{\text{total}} = 8$ slots for the inference period (i.e. $N_{\text{total}} - N_{\text{train}}$ inference period).

Fig. 3 illustrates the FER performance comparison among different channel estimation methods. The proposed Wiener filter method significantly outperforms both the conventional and data-aided CE methods across all SNR regimes. Moreover, the performance of the proposed method using online training (*Proposed (Online)*) closely approaches that of the ideal case where true channel information is available for training (*Proposed (True)*), demonstrating the effectiveness of our online data generation strategy. This result implies that utilizing data-aided CFR estimates as substitutes for true CFRs during the generation of online training data is effective for determining the adaptive Wiener filter. Consequently, this approach facilitates noise removal from the estimated channel, even in the absence of prior knowledge regarding channel and error statistics.

Fig. 4 shows the NMSE performance of the proposed Wiener filter method for different window sizes (W_t, W_f). NMSE is used to isolate the effect of window size on the accuracy of channel estimation, independent of the system-level coding and detection process. When the window size is small, the denoising performance is limited due to the lack of sufficient information from neighboring CFR estimates. As the window size increases, the NMSE gradually improves because the filter can better exploit the time-frequency correlations. However,

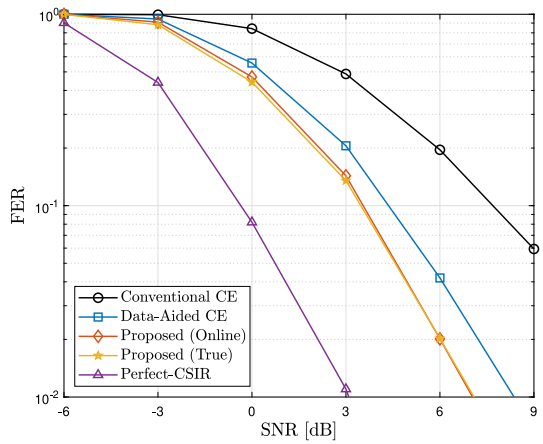

Fig. 3. FER comparison between the proposed and other channel estimation methods. Window size $(W_t, W_f) = (3, 3)$ is used for the proposed method.

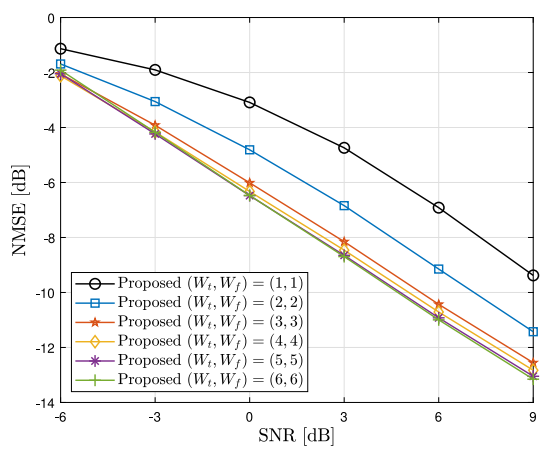

Fig. 4. NMSE comparison between the proposed method with varying time-frequency window sizes (W_t, W_f).

Table 1
NMSE comparison (in dB) across different N_{train} and SNR levels.

N_{train}\SNR	−3	0	3	6
2	−3.878	−5.960	−8.080	−10.380
3	−3.880	−5.960	−8.095	−10.384
4	−3.882	−5.964	−8.112	−10.387
5	−3.890	−5.979	−8.125	−10.399

the performance gains saturate around $(W_t, W_f) = (3, 3)$, indicating that additional window expansion beyond this point offers negligible benefits. This saturation occurs because the channel correlation in the time-frequency domain becomes weak beyond a certain range, limiting the effectiveness of incorporating more distant CFR estimates.

Table 1 presents the NMSE performance of the proposed method as a function of the number of online training slots N_{train}. As N_{train} increases, the NMSE improves slightly, indicating that a larger training set provides better estimates of the cross-covariance vector $\boldsymbol{\theta}$ in (17) and the auto-covariance matrix $\boldsymbol{\Phi}$ in (18). However, the improvement is marginal, suggesting that the proposed method can achieve robust performance even with a small amount of online training data (i.e., a short training period).

Table 2 compares FER across various window sizes and APP thresholds at 0 dB and 9 dB SNR levels. At 0 dB, a high APP threshold ($\tau_{\text{APP}} = 0.8$) yields the best FER for most window sizes, except $(Q_t, Q_f) = (1, 1)$, where using all detections ($\tau_{\text{APP}} = 0$) performs better due to noise averaging. At 9 dB, lowering the APP threshold benefits larger

Table 2
FER comparison across different window sizes (Q_t, Q_f) and APP thresholds (τ_{APP}) for online training.

(a) SNR = 0 dB [All values × 10^{-1}]					
$(Q_t, Q_f)\backslash\tau_{\text{APP}}$	0	0.2	0.4	0.6	0.8
(1,1)	**5.048**	5.173	5.141	5.110	5.128
(2,2)	4.706	4.689	4.684	4.642	**4.636**
(3,3)	4.589	4.584	4.557	4.531	**4.521**
(4,4)	4.552	4.553	4.534	4.509	**4.490**
(5,5)	4.584	4.593	4.572	4.529	**4.508**
(b) SNR = 9 dB [All values × 10^{-3}]					
$(Q_t, Q_f)\backslash\tau_{\text{APP}}$	0	0.2	0.4	0.6	0.8
(1,1)	2.838	2.813	2.525	**2.425**	2.438
(2,2)	2.688	2.825	2.663	**2.600**	2.738
(3,3)	2.775	2.825	**2.538**	2.613	2.725
(4,4)	2.796	**2.663**	2.850	2.713	2.788
(5,5)	3.033	**2.650**	2.850	2.675	2.988

windows, as symbol reliability is high and more virtual pilots enhance estimation. However, overly large windows can still degrade FER, even at high SNR, due to reduced local stationarity and overfitting to weakly correlated pilots. These results emphasize the need to adaptively adjust the APP threshold and window size based on SNR and channel dynamics.

5. Conclusions

In this paper, we proposed an online adaptive channel denoising approach for MIMO-OFDM systems operating over doubly-selective channels. To enable online adaptation, we developed a novel training data generation strategy that leverages data-aided channel estimates as practical substitutes for the unavailable true CFRs. Using the generated online dataset, the denoising filter taps were optimized based on the Wiener filter principle. Extensive simulation results demonstrated that the proposed approach significantly improves channel estimation performance over conventional methods, while exhibiting strong adaptability and efficiency across diverse channel conditions.

CRediT authorship contribution statement

Sungyoung Ha: Writing – original draft, Validation, Software, Investigation. **Anseok Lee:** Methodology, Investigation, Conceptualization. **Heesoo Lee:** Supervision, Resources, Project administration. **Yo-Seb Jeon:** Writing – original draft, Supervision, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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